

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
December 2024	All	All	CA&AS	Reinstated – updating in process
October 2024	All	All	CA&AS	Audit program discontinued – kept for reference use only
September 2023	Key Control Question: 1. Environmental Management Programs Primary Test: Energy Management	Approach Control Assessment Questions Added Environmental Evaluation and Use of Approved Suppliers	Priti Kadam	Changes were identified as needed during our annual review of Corporate Environment Affairs Professional Self-Assessment (PSA) program.
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

This audit program identifies various types of activities that are categorized as environmental management. They should be present at IBM Chemical using MDR (manufacturing, development, and research) locations. These activities are in the areas of environmental management, chemical management, and energy management. IBM is committed to reducing the environmental impact of all of its activities, including the efficient use of energy.

Environmental management, chemical management, and energy management at headquarters, sales locations, offices, assembly, and light manufacturing sites are audited as part of other audit programs, such as Real Estate Strategy and Operations (48-10-00) and Manufacturing (54-00-00).

- The environmental management process rating includes all of the environmental management and chemical management programs tests.

The environmental management system (EMS) program at each site will be included in the testing for the environmental management and chemical management processes. Implementation of corrective action is required for each non-conformance and follow-up by audit will be performed to confirm closure of each non-conformance.

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PLANNING: Audit Program Risk Alignment

Name of Process

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Incident Reporting and Prevention	High	Environmental Management Programs	EM005.P
Use of Approved Suppliers	High	Environmental Management Programs	EM007.P
Regulatory Compliance	High	Environmental Management Programs	EM003.P, EM005.P, EM008.P, EM009.P, EM010.P, EM011.P, EM012.P, EM013.P, EM014.P, EM019.P, EM022.P
Measurement, Monitoring, Reporting, and Management Review	High	Environmental Management Programs	EM002.P, EM004.P, EM005.P, EM010.P, EM011.P, EM012.P, EM013.P, EM014.P, EM015.P, EM016.P, EM021.P, EM022.P, EM023.P
Staffing	Medium	Environmental Management Programs	EM001.P
Impact Assessment	Medium	Environmental Management Programs	EM006.P, EM020.P
Energy Management	Medium	Environmental Management Programs	EM004.P
Training and Communications, Management of On-Site Contractors	Medium	Environmental Management Programs	EM017.P, EM018.P
Evaluation of Tenants	Low	Environmental Management Programs	EM019.P, EM020.P

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PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Analysis to be Performed	Concern
Chemical Management	• Validate appropriate reviews and approvals have been completed for on-site chemicals (IBM and contractor). • Review a selected sample of chemical purchase orders to validate proper approvals. • Verify completeness of superfund amendments and re-authorization act (SARA) reports, approved/reviewed by management prior to submission, and completed in a timely manner. • Core technology development chemicals (CTDC) were: • Reported to authorized department managers • Industrial hygiene review of CTDC characteristics • Review SOD and COInS access.	• Unauthorized purchase, management, and use of chemicals by employees and contractors at the site may increase risk of safety concerns and lead to injury or damage to employees and the environment. • Inadequate container and pipe labeling may provide inaccurate hazard information to employees and increase chances of injury. • Inadequate communication of chemical hazards could lead to employee injury. • Inadequate inventory practices may cause operational deficiencies and lead to the mismanagement of hazardous substances and misrepresentation of reports to regulatory agencies.

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Concern
Chemical Management (continued)	• Select a sample of IBM/contractor/tenant chemicals and perform forward and reverse checks to validate: ○ Chemicals found within departments were appropriately authorized. ○ SDSs were up to date and accurate. ○ Containers were properly labeled. • Determine if employees were aware of how and when to obtain SDSs. • Review results of inventory activity for completeness and activities performed to resolve potential discrepancies (chemical distribution center [(CDC])). • Review inventory activity for independent reconciliation and management awareness.	

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Concern
Hazardous Waste	- Review a sample of hazardous waste collection and storage sites and determine if they are handled and stored within guidelines. - Review waste aging and areas of storage to determine if practices are within permit parameters and accumulation. - Review the hazardous waste storage permit renewal. - Review a sample of hazardous waste shipments to determine if: - Shipments were processed in accordance with governmental and corporate requirements related to the shipment of hazardous materials. - Receiving, shipping, and transferring documentation were completed, accurate and shipped to an IBM approved waste vendor. - Employees with access to hazardous materials were trained as required.	- Hazardous waste collected and stored inappropriately can have a negative environmental impact and may result in permit violations, regulatory fines, and increased costs. - Unsafe handling of hazardous materials could increase the potential of employee exposure and injury. - Incomplete or missing documentation could result in the failure to demonstrate compliance with shipping regulations and lead to regulatory fines. - Employees that are not trained and not authorized to handle hazardous materials could result in environmental issues as well as employee illness and injury.
Water Discharge	- Review a judgmental sample of documentation submitted to city, state, or federal regulatory agencies to determine if: - Reports were prepared and submitted on time and accurately. - Valid operating permits or licenses were in place. - Review the Water Discharge process to verify: - Compliance with CI ENV 204--water discharges. - Training requirements had been met in a timely manner.	Lack of monitoring of the contaminate levels in the water discharge to verify compliance with regulations could result in monetary fines, negatively impact environmental surroundings and IBM's image.

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Concern
Management of On-site Environmental Contractors	• Review a judgmental sample of contractor deliverables related to water discharge, hazardous waste, and chemical management to determine if: • Contractor had performed statement of work deliverables, completely, timely and accurately. • There was a documented process to communicate IBM requirements to contractors. • Procedures had been established for the on-going monitoring of contractor to confirm adherence to IBM requirements. • A formal communication process had been established to document vendor deficiencies in the performance of the scope of work to IBM requirements, and corrective action plans had been implemented as appropriate.	• Lack of IBM oversight of on-site environmental contractors could result in monetary fines, negatively impact environmental surroundings and IBM's image.
Permits	• Review all site permits to determine if the permits were valid.	• Lapses in permit validity could result in regulatory fines and impact site operations and damage IBM's image.
Environmental Incident Reporting	• Review all recorded incidents to ensure: • Accurate, complete and timely records. • Compliance with external agencies and internal management requirements. • Root cause analysis was performed.	• Failure to monitor, analyze and report an incident could lead to legal penalties and damage to the environment.
Spill Prevention	• Review Spill Control and Countermeasure Plan (SCCP) to confirm an appropriate containment plan is in place. • Select a sample to validate: • Tank leak check reviews were performed correctly. • Containment reviews and analysis were completed by the facilities management vendor. • Perform walk-through to inspect tanks and containment areas.	• Failure to implement and follow a spill prevention control plan could lead to unnecessary environmental incidents, damage to the environment, and legal ramifications.

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Concern
Management of On-site Environmental Contractors	• Review a judgmental sample of contractor deliverables related to water discharge, hazardous waste, and chemical management to determine if: • Contractor had performed statement of work deliverables, completely, timely and accurately. • There was a documented process to communicate IBM requirements to contractors. • Procedures had been established for the on-going monitoring of contractor to confirm adherence to IBM requirements. • A formal communication process had been established to document vendor deficiencies in the performance of the scope of work to IBM requirements, and corrective action plans had been implemented as appropriate.	• Lack of IBM oversight of on-site environmental contractors could result in monetary fines, negatively impact environmental surroundings and IBM's image.
Energy Usage and Conservation	• Review the measurements to ensure that the energy conservation targets have been met by the location.	• Not meeting the energy conservation targets set by the company and local governments could lead to potential fines to IBM.
Non-Hazardous Waste	• Review the measurements to ensure that the recycling targets have been met.	• Not meeting the recycling targets set by the company and local governments could lead to potential fines to IBM.

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PLANNING: Reference Material

More specific references are listed with each test. This list includes the primary references and associated intranet addresses.

- Corporate Policy 139, Environmental Affairs
<http://w3-03.ibm.com/ibm/documents/corpdocweb.nsf/ContentDocsByTitle/Corporate+Policy+139>
- Corporate Instructions, Environmental Affairs and Product Safety
<https://w3.ibm.com/w3publisher/corporate-environmental-affairs/about/corporate-policy-and-instructions>
- ENV 205 Approved Category III Suppliers
<https://w3.ibm.com/w3publisher/corporate-environmental-affairs/programs/supplier-evaluation-and-approval>
- IBM Worldwide Environmental Management System Manual
<https://w3.ibm.com/w3publisher/corporate-environmental-affairs/ems/documents-related-to-global-ems>
- General Data Protection Regulation (GDPR)
GDPR Readiness: <https://www.ibm.com/data-responsibility/gdpr/#4>
Compliance, Audit, Risk Mitigation & Management: <https://w3.ibm.com/w3publisher/dprc>

Glossary

It is expected that auditors will familiarize themselves with various terms related to environmental affairs during initial tours of the location and in discussions with location management. The following definitions of the terms used in this program are provided to assist the auditor in gaining an understanding of the key issues to be considered. These definitions may differ from those used by regulators and agencies outside IBM. Each IBM location is responsible for understanding the specific criteria, regulations, and requirements applicable to the location.

Chlorofluorocarbons (CFCs). A family of inert and relatively non-toxic chemicals used in refrigeration, packaging, and insulation or as solvents and aerosol propellants. CFCs are not destroyed in the lower atmosphere and drift into the upper atmosphere where their chlorine components destroy the ozone.

Environmental Affairs Document of Understanding (DOU). A formal agreement establishing environmental affairs responsibilities and signed between affected parties, including the designated division environmental affairs focal point(s), the site operations manager, the site senior location manager, and the country environmental affairs focal point.

Effluent. Wastewater treated or untreated, that flows out of a treatment plant, sewer, or industrial outfall.

Effluent Limitation. Restrictions established by a government agency on quantities, rates, and concentrations in wastewater discharges.

EIA - Environmental Impact Assessment, used to evaluate site processes and their potential impacts to the environment.

EIRS - Environmental incident reporting system, an application used by IBM for the reporting and tracking of unauthorized chemical releases and other environmental incidents.

Emission. Pollution discharged into the atmosphere from exhaust stacks, vents, and surface areas of commercial or industrial facilities.

Emission Limitation. The maximum amount of air emissions legally allowed from a source.

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PLANNING: Reference Material (Continued)

Emission Unit. A single or group of emission points discharging into the atmosphere.

Environmental Management System. IBM's worldwide environmental management system as issued by IBM Corporate Environmental Affairs. IBM's guide for the implementation of an environmental management system, and for the implementation of an ISO-14001 environmental management system.

Exceedance. An emission or effluent that exceeds allowable effluent limitations, emissions standards, allowable limits, or concentration levels. Permit exceedances must be reported annually to Corporate Environmental Programs in the Environmental Master Plan.

Exposure Monitoring. A monitoring process to evaluate potential exposure to hazardous materials, including mists, vapors, dusts, and gases. Samples are conducted in areas or in employee breathing zones to test for actual exposure to these materials.

Groundwater. The supply of water found beneath the earth's surface. Groundwater is a major source of drinking water in some areas.

Hazardous Waste. Materials that have been designated by federal, state, or local governments as hazardous waste.

HAZCOM. Hazard communication, a requirement applicable to certain U.S. locations.

International Chemical Data System (ICDS). This is a central database of material safety data sheets for chemicals authorized for internal use as well as for IBM products (field use materials [FUMs]).

Incident. Is one of the following: a government action that results in a fine, penalty, or adverse publicity; a noncompliance; a release.

Influent. Water, wastewater, or other liquid flowing into a reservoir, basin, or treatment plant.

Local Exhaust Ventilation Program. A program that specifies controls to determine proper design, operation, maintenance, installation, and use of local exhaust systems that are installed to protect an employee's health and safety.

Methyl Chloroform (MCF). IBM no longer uses MCF in its products and processes (see CFCs) except in oxidation furnaces where MCF is virtually destroyed.

Monitoring. Periodic or continuous surveillance or testing to determine the pollutant levels in various media (for example, air, wastewater, and water).

Nonhazardous Waste. Materials that have not been designated as hazardous by federal, state, or local governments. Examples would be sanitary sludges, cafeteria waste, paper, cardboard, plastics, monochrome cathode ray tubes, and wood.

Outfall. The place where an effluent is discharged into receiving waters.

Product Environmental Profile. The product environmental profile program documents the environmental characteristics of IBM products (in the development stages) and identifies potentially hazardous components, parts, and assemblies which may require special handling or disposal at product end-of-life. PEP documentation is also intended to be used as an information source to respond to customer inquiries concerning the chemical and environmental attributes of IBM products.

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PLANNING: Reference Material (Continued)

Permit. An authorization, license, or equivalent control document issued by a government agency to implement the requirements of an environmental regulation (example, a permit to operate a wastewater treatment plant or to operate a facility that may generate emissions).

Primary Containment. A device such as a tank, pipe, and drum which holds industrial liquids or chemicals.

Release (IBM). A continuous or instantaneous, unauthorized discharge of a chemical substance. A release includes any pouring, emitting, emptying, discharging, dumping, injecting, escaping, pumping, leaching, spilling, or disposing from the primary containment.

Reportable Release (IBM). A release from primary containment that is reportable under the requirements of Corporate Instruction. These must be reported to Corporate Environmental Programs.

Respiratory Protection Program. A program that details the proper selection, inspection, maintenance, and use of appropriate respiratory protection used for protecting employees' health primarily on a short-term basis during non-emergency situations. It consists of respirator selection criteria, audits, medical and industrial hygiene evaluation, and approval and record maintenance.

RCRA. U.S. Resource Conservation and Recovery Act.

SARA. U.S. Superfund Amendments and Reauthorization Act.

Sanitary Sewer. Underground pipes that carry off domestic waste which could include industrial waste, but not storm water.

Secondary Containment. A system which will completely collect and contain all primary containment spills, leaks, and contaminated precipitation until appropriate remedial action can be taken. Control features for containment may include leak detection such as liquid or level sensors, level monitoring, chemical sensors, and alarms.

Solid Waste. As used by IBM, refers to non-hazardous solid waste that would typically be disposed of in a sanitary landfill. Examples are wood pallets, paper, cardboard, newspapers, and cafeteria waste.

Storm Sewer. A system of pipes (separate from sanitary sewer) that carries only stormwater runoff from building and land surfaces.

Surface Water. All water naturally open to the atmosphere (rivers, lakes, reservoirs, streams, impoundments, seas, and estuaries) and all springs, wells, or other collectors which are directly influenced by surface water.

TSCA. U.S. Toxic Substance Control Act.

Waste Stream. Unwanted materials such as wastewater, waste chemicals, and waste materials left over from a manufacturing process.

Wastewater. The spent or used water from human activities or industrial processes.

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Illness/Accident/injury/incident reporting	Review and approval of required forms to record the situations and report to the corresponding agencies	Request and review all of the documentation required by the process to record and report all incidents
Documentation	Review and approval of documentation required to record and note physical findings as tested in the process	Verify that documentation is consistent with physical findings and that approvals and paperwork do not appear to be altered
Bribes	Review and approval process for contracts and vendors. Accounts payable and WWERS verification of payments.	Review all documentation, licenses, and government approvals; verify that IBM was qualified for the license and approval. Pull and review a sample of payments corresponding to the licenses and approvals.
Chemical Inventory	Reconciliation of stored inventory to receipts, and requested/used inventory. Inventory SOD	Review inventory or inventory changes done repetitively by the same person (s)
Procurement	Reviews and testing of procurement process including POs and actual orders.	Test to ensure that the materials that we have received are in fact what is listed on the PO and that it is required and has been ordered by the appropriate personnel

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Objective:

Verify that environmental affairs programs and management controls were effectively implemented as required by IBM and Legal.

Risk/Exposure:

Lack of environmental programs and management controls could lead to noncompliance with IBM, legal, and/or environmental management system (EMS) requirements resulting in a threat to employee safety, the environment, and potential legal action against IBM. In addition, undetected or inadequately handled releases could result in expensive corrective actions and a negative impact to IBM's image.

Absence of an environmental impact assessment could result in a lack of information necessary for the effective and timely identification of potentially adverse environmental effects of IBM manufacturing and facilities processes.

Absence of a product environmental profile could result in a lack of information necessary to identify the environmental characteristics of IBM products, and to identify potentially hazardous components, parts, and assemblies which may require special handling or disposal at product end-of-life.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Staffing (EM001.P)

Objective:

To verify that the roles and responsibilities for implementing IBM's environmental management system (EMS) have been identified and staffed by qualified personnel.

Approach:

Review organization charts, procedure(s), documents of understanding where they exist, job descriptions, and experience for personnel responsible for executing environmental programs. Validate that the following controls had been implemented:

1. An appropriate level of management has been identified and designated to provide a coordinated focus on environmental and chemical management requirements, and to lead implementation of IBM's environmental policy, programs and environmental management system for the location. Verify this has been communicated within the organization and to the affected personnel.
2. The location has identified an environmental person who is responsible for developing and maintaining environmental programs.
3. Environmental programs are staffed and monitored by qualified personnel who are technically competent to evaluate and ensure IBM's compliance with applicable legal and IBM requirements.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Management System (EM002.P)

Objective:

Verify the location has established and is maintaining its EMS in line with IBM requirements. Determine that the location has implemented a procedure for the identification of significant environmental aspects and establishing site-specific objectives and targets. Determine that consideration was given to setting location specific objectives and targets, in addition to corporate level objectives and targets, to support IBM policy objectives for continual improvement of EMS and environmental performance. This process shall consider the context, risks and opportunities and relevant requirements of relevant interested parties.

Approach:

Review the EMS and:

1. Verify that an EMS has been established and implemented in line with IBM Worldwide EMS Manual requirements, and other requirements, including CI ENV 201.
2. Verify that the documents and procedures that are required to implement EMS requirements and execute IBM's environmental policy, instructions, and other requirements of IBM's EMS are in place, current, controlled, and communicated. Verify that a process is in place which requires periodic review of Environmental Affairs procedures to assess if they remain relevant and current.
3. Verify that the location is complying with records retention requirements for Environmental Affairs documents. Verify that documents, regulatory reports, and back-up documentation, subject to records retention requirements, are available in the environmental filing system in accordance with the requirements.
4. Verify that roles and responsibilities have been clearly defined and communicated to affected business functions and personnel. Where appropriate, verify that Documents of Understanding defining program responsibilities shared by multiple organizations or other means of communication with supporting organizations, e.g., Procurement, Global Logistics, Global Real Estate (GRE), etc., have been approved, signed, and implemented, as required.
5. Verify that a procedure is in place to make employees and contractors on IBM premises aware of IBM's environmental policy and follow it as it applies to their job responsibilities. Verify a documented process is in place to communicate IBM's requirements and changes to the requirements to on-site contractors and subcontractors who provide environmental and chemical management services.
6. In environmental roles where specific training requirements are prescribed by legal requirements, verify that such requirements are being met.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Management System (EM002.P)

(Continued)

7. Review the process for the identification of environmental aspects to verify the following considerations:
 - a. The process considers the context, risks, opportunities, and relevant requirements of relevant interested parties.
 - b. The procedure used for identifying the significant environmental aspects, defined in the Worldwide EMS Manual, has been implemented.
 - c. The rationale used for determining significant aspects considers criteria established in the procedure.
 - d. The significant aspects identified at the corporate level have been identified at the location, where applicable.
 - e. The locations maintain a list of significant environmental aspects and procedures are in place to maintain operational control of the identified significant environmental aspects.

Guide Note: this can be verified during other tests in this audit program.

8. Review the goals for environmental sustainability established for the locations' EMS for the following:
 - a. Verify that the location's EMS supports the goals set at the Corporate level and/or by the location.
 - b. Review the location's EMS and validate that it is current and being followed and that it includes the following with respect to achieving the goals:
 - i. Responsibility for meeting each goal has been identified and assigned.
 - ii. Performance is monitored against established goals.
 - iii. The means for achieving goals.
 - iv. Results are periodically reviewed with appropriate levels of management.
 - c. Validate that the program is reviewed annually.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: IBM Compliance Obligations (Legal and Other Requirements), Inspections, and Regulatory Commitments (EM003.P)

Objective:

To verify that an effective process exists for meeting IBM Compliance Obligations including IBM, legal, and permit requirements.

Approach:

1. Verify that the location has a process by which IBM's Compliance Obligations (legal, IBM and other requirements) and their changes, proposed or promulgated, are tracked, analyzed, reported to management, and that compliance plans for applicable requirements have been executed.
2. Verify that a procedure is in place to identify and provide access to applicable IBM Compliance Obligations, (legal, IBM and other requirements) to which IBM and/or the location subscribes to affected personnel. Determine if the location has a record of legal permits that have been issued or for which have been applied. Verify that there is a process to determine that permits are obtained, renewed, or modified in a timely manner.

Guide Note: Testing that the location follows their permit requirements and verification of the permit processes for hazardous and non-hazardous waste streams, air emissions, industrial wastewater discharge, sanitary wastewater discharge, storm water runoff, and groundwater will be conducted as part of the individual program tests for these areas.

3. Evaluate the location's documented permit management process and verify that:
 - a. An individual has been assigned the responsibility for oversight of the permit process.
 - b. A process has been implemented to assure that permits are obtained, renewed, or modified as required. Select a representative sample of filings and verify that all required filings were reviewed by appropriate levels of management and submitted as required.
 - c. An inventory of permits, including application and renewal dates, has been established and that this inventory has been reviewed and updated once a year.
 - d. A process exists and is effectively executed to communicate permit status to appropriate management.
4. Test the location's environmental permits to verify that they are current and the requirements for discharge limits, measurements, reporting, fees, and renewal dates have been identified.
5. Request a list of new regulations proposed or promulgated during the audit period and verify that an impact statement has been reviewed by the management team, and if any issues are identified, prepare an action plan to address them. The impact statement should contain a review of new regulations proposed or promulgated during the audit period and their associated impacts with action plans.
6. Request a list of all relevant regulatory and legislative requirements for the jurisdictions in which the facility is located and the system by which the facility keeps this current. Based on available government or other resource databases identified, verify that the list of applicable requirements is complete. Using this list as a basis, select a representative sample of regulatory reports and verify coverage is complete.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: IBM Compliance Obligations (Legal and Other Requirements), Inspections, and Regulatory Commitments (EM003.P)

(Continued)

7. Verify that the location has a process to document environmental regulatory inspections and verify that the location has a process to report any findings, notices of violation, or fines to the Environmental Incident Reporting System (EIRS) and that action plans are in place to close any findings, notices of violation, or fines. Verify that the location has complied with all required commitments resulting from inspection findings and incident reviews or that alternate compliance schedules have been arranged with management and the controlling government agency or entity.
8. Verify that the location has a procedure to track all open action items or commitments made to legal authorities through the legal permitting, reporting, or inspection process, and those made to legal authorities through voluntary notification and/or location management or Corporate Audit through self-audit or corporate audit processes.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Energy Management (EM004.P)

Objective:

To verify that the location has an effective energy management program in place, in conformance with Corporate Instruction (CI) ENV 206 to control the use and cost of energy through implementation of energy conservation projects. To determine if the location's established energy conservation objectives and targets support the environmental sustainability goals established by corporate.

Approach:

Review the site energy management program and validate that the following attributes are functioning effectively:

1. Verify the location has a designated Energy Management Program with the following attributes, as required by CI ENV 206, are in place:
 - a. A designated individual to oversee the IBM location's Energy Management program. A single individual may oversee multiple IBM locations and a documented Energy Management plan.
 - b. Defining and communicating Roles and Responsibilities - Have roles and responsibilities for execution of EnMP requirements been defined and communicated.
 - c. Where responsibilities relating to the Energy Management program are shared between organizations or delegated by one organization to another organization, ensure Documents of Understanding (DOU) or other communications which define the program responsibilities between affected organizations have been clearly documented and approved, established and implemented, as required, or that responsibilities are clearly documented in their respective environmental management systems.
 - d. Is management maintaining the relevant skills of the energy management personnel?
 - e. A two-year (current + 1) energy conservation work plan detailing planned capital and expense energy conservation projects to support IBM's energy conservation goal.
 - f. The location has a business process to accurately collect and report utility billing and energy conservation project information to Envizi.
 - g. For the Top 10 IBM energy consumption locations verify the energy management program has been reviewed with a documented approval as specified in CI ENV 206.
 - h. Procurement of energy consuming equipment - When the forecasted energy consumption of the purchased equipment at a Tier 1 location is greater than 1% of the location's energy consumption, Tier 1 locations shall inform suppliers that energy performance is one of the evaluation criteria for procurement. Review all purchases of equipment that meet this condition and ensure that such language is included in contracts with providers that may execute these purchases on IBM's behalf

Guide Note: This question is intended to verify that all the energy management plan requirements detailed in CI ENV 206 have been completed and documented at the location.

2. Verify that the locations with greater than 2,000 MWH/year or more of total energy consumption, have completed the applicable GRE Energy Conservation Best Practices SURVEYS in Envizi and that the location has budgeted the capital and expense necessary to achieve the energy conservation savings projected in its plan for the current year.
 - a. Request a list of energy conservation projects that have been completed in the past 12 months. Select a sample of the projects and verify the following:
 - i. That the savings calculations can be substantiated either by direct metering before and after the energy efficiency improvement, or by supportable calculation.

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- ii. That the project was implemented as described in the energy management program.
- iii. That the energy conservation entry in Envizi matches the energy conservation calculation in the site file. Where projects are combined for Envizi reporting, verify that the sum of the projects reported in Envizi match the sum of the individual projects in the site records.
- iv. For all projects with a value of over \$500,000 capital and expense, verify that the CI ENV 206 Appendix C, Energy Efficient Design Practices was used in the design and that any exceptions were documented in the process. Select a representative sample for review.

Guide Note: The audit should verify that the selected project(s) used the specified, energy-efficient systems for the building envelope, chillers, lighting, motors, uninterruptible power supplies (UPSs), and other facility infrastructure equipment or provided a justification for their exclusion from the project included in the original questions above.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Energy Management (EM004.P)

(Continued)

3. Verify that the location has a process to track and identify relevant energy and greenhouse gas (GHG) emissions reporting requirements. Request a list of all legal requirements that are applicable to the energy program, which may include greenhouse gas emissions reporting, and verify that the necessary processes are in place and operational to ensure compliance with requirements, collect necessary data, and make necessary reports.
4. For Tier 1 locations verify the requirements of IBM's Worldwide Energy Management Program (WWEEnMP), which are set out in Addendum A and B of the WWEEnMP are fully executed and are incorporated within the location's Environmental Management System documentation and that the internal EMS audit of the Energy Management Program has been conducted on the required schedule.

Guide Note: This question is only applicable to Tier 1 locations (locations consuming greater than 12,000 mWh in a year).

URL Name: EnMP Requirements

URL: <https://w3.ibm.com/w3publisher/corporate-environmental-affairs/ems/documents-related-to-global-ems>

5. Has the Site or the Business Organization a process in place to identify existing and up-coming opportunities for the procurement of Contracted Renewable Energy, in line with attaining linear progress toward IBM's corporate goal, and is that process being executed as planned?

Has the Site or the Business Organization a plan in place to execute on identified opportunities for the procurement of Contracted Renewable Energy, in line with attaining linear progress toward IBM's corporate goal?

Did the Site or the Business Organization achieve "on-track" or "green" status for the last two quarters of the previous year in Corporate Environmental Affairs' quarterly internal "Scorecard for IBM's Environmental Sustainability Goals"?

Guidance:

Every quarter, Corporate Environmental Affairs publishes the above-mentioned scorecard and rates performance of Business Organizations as "on track", "behind", or "off track", based on clearly defined criteria. This is a document for IBM Internal Use Only and is distributed to the executives responsible for overseeing execution against the goals and the individuals directly responsible for execution. If the answer to the above question is NO, then review what actions is the Site or Business Organization going to implement so that it achieves "on-track" or "green" status for the last two quarters of the current and following years.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Incident Prevention and Reporting (EM005.P)

Objective:

To verify that the location has a documented process in place for the prevention, control, and response to environmental incidents per IBM and legal requirements. In addition, verify that environmental incidents are reported and tracked as required by IBM and legal requirements.

Approach:

1. Verify that the location has a documented environmental incident prevention and reporting program and that a process is in place requiring a periodic review of the program for necessary updates per legal and IBM requirements at least once a year. Verify that legal requirements, including recent changes have been incorporated into the incident prevention and reporting program and that all commitments made in the spill prevention and response plan have been met or that action plans are in process.
2. Review the location's documented environmental incident prevention and reporting process. Ensure that this has been reviewed and updated at least once a year.
3. Determine if areas of potential releases, including chemical and waste storage areas, have been identified. Inspect a sample of these areas to validate that they have been constructed in conformance with IBM CI ENV 204 Attachment E and legal requirements.

Guide Note: CI ENV 204 Attachment E -- Requirements for Liquid Storage and Handling Systems

<https://w3.ibm.com/w3publisher/corporate-documents/env/env-204-attachment-e>

4. Validate that the location has an effective emergency response plan for unplanned chemical releases and verify that periodic training occurs for the incident responders to assure their knowledge is current and they have an ongoing familiarity with reporting requirements.
5. Verify the spill prevention and emergency response plan has been updated and tested per legal and IBM requirements at least once a year.
6. Verify that periodic training occurs for the incident responders to assure their knowledge is current and have an ongoing familiarity with reporting requirements.
7. Validate that the location has a process to track environmental incidents.
8. Select a sample of environmental incident reports (EIRS) for the audit period and test the following:
 - a. The reporting matrix was in place, adhered to, and decisions were documented.
 - b. Incidents meeting the reporting criteria in CI ENV 204 were reported accurately.
 - c. The location's incident response team responded in an appropriate and timely manner.
 - d. Reviews of incidents were conducted with appropriate levels of management.
9. Verify that there is a process to track and analyze environmental incidents to prevent their recurrence and to implement corrective actions to prevent future releases. Corrective actions should include target completion dates and tracking to closure.
10. Verify the records are retained per IBM and legal requirements.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Impact Assessment (EM006.P)

Objective:

To verify that environmental impact assessments (EIAs) are prepared to identify and prevent adverse effects on the environment from manufacturing activities and major facility infrastructure processes at IBM, and from manufacturing activities performed by contract manufacturers for IBM. Verify that EIAs are in compliance with IBM requirements.

Approach:

1. Verify that a documented process exists for preparing, updating, and reviewing EIAs for development, manufacturing (IBM and contract manufacturers), and facilities operations. Verify that the location process covers all the requirements of CI ENV 204.
2. Verify that Process EIAs were conducted during the design phase and completed and approved prior to introduction of a new or modified process. Review a representative sample of all EIAs conducted during the audit period.

Guide Note: CI ENV 204: Identify the total number of new or modified processes requiring an individual Process EIA during the audit period. Also, include:

- a. A list of new and modified manufacturing or development process, and
- b. A list of new and modified facilities infrastructure process.

Critical pieces of data should include contents of EIA per CI ENV204. Verification should include whether:

1. EIA reviews took place prior to the operation of the process
 2. EIA reviews did not occur for some projects
 3. EIA reviews were not completed in a timely manner to allow identification and resolution of issues
 4. EIA reviews were not consistently executed to location and IBM requirements
 5. EIA reviews were not consistently performed for newly installed or existing processes to assure data accuracy
 6. Identified issues from the EIA review were not tracked and closed.
3. Obtain a listing of facilities and production equipment capital or installation projects initiated during the audit period. Select a representative sample of EIAs for new facilities projects and test the following attributes:
 - a. Projects were reviewed for environmental, health and safety impacts.
 - b. Necessary legal reports, permit applications, and permits were completed per commitment.
 - c. Review was complete with regard to the requirements of the controlling location document for EIAs and CI ENV 204.
 - d. Status of all requirements created during the process EIAs, owners, closure dates and status.
 - e. If action plans were created as part of the reviews, verify owners were identified, dates assigned for closure and status monitored.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Impact Assessment (EM006.P)

(Continued)

4. Select a representative sample of EIAs for processes transferred to other IBM locations, or to a contract manufacturer where the contract manufacturer was contractually obligated to use a process developed by IBM, or processes which IBM specifies manufacturing equipment or procedures unique to the contract manufacturer, or IBM furnishes chemicals or chemical using equipment. Test the following:
 - a. EIAs were prepared prior to the internal transfer or implementation of the process at a contract manufacturer.
 - b. IBM banned or restricted chemicals were not specified in the process or contract.
 - c. Significant changes to the original process were identified and EIAs updated, including changes to the process' original purpose, chemical use, major equipment, utilities, or engineering controls for environmental or employee protection.
5. For major manufacturing locations, verify that process EIAs are completed and maintained on the Notes Tools Connect Process EIA System or alternate reporting methods according to requirements in CI ENV 204.
6. Verify that new, amended, or renewed lease agreements with tenants that conduct manufacturing, hardware development, or chemicals-using research operations contain provisions requiring the tenant to provide IBM with process EIA information, prohibiting discharge of chemicals into receiving water / wastewater that form perfluorooctyl sulfonates (PFOS) or perfluorooctanoic acids (PFOA, and prohibiting use of IBM banned or restricted chemicals.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Evaluation and Use of Approved Suppliers (EM007.P)

Objective:

To validate that a process exists to verify that suppliers are able to operate in compliance with IBM and legal requirements.

Approach:

Select a representative sample of category III suppliers providing hazardous waste disposal, special waste disposal, and product end-of-life management, Extended Producer Responsibility (EPR) systems, and category II suppliers to whom IBM specifies and/or furnishes chemicals, process equipment, or contaminated equipment, and verify they have been reviewed and approved per the requirements of CI ENV 205. At the site, testing should include only locally procured suppliers, not those included in a national contract owned by another site.

1. Verify the location has a process to identify suppliers providing services providing their services on non-IBM location and which may need CI ENV 205 evaluation.
2. Verify that suppliers providing hazardous waste or nonhazardous special waste treatment and/or disposal services (Category III) have been approved by Corporate Environmental Affairs (CEA) prior to entering a contract with IBM.
3. Verify that suppliers providing product end-of-life management services, including EPR systems (Category III) have been approved by CEA prior to entering a contract with IBM or IBM's global logistics service provider.
4. Verify that suppliers which IBM specifies and/or furnishes chemicals, process equipment, or contaminated equipment (Category II) have been reviewed and approved per CI ENV 205 requirements.
Select a representative sample of suppliers and verify that all hazardous waste, nonhazardous special waste, and end-of-life products, as defined in CI ENV 204, have been shipped to IBM approved suppliers or managed by IBM-approved EPR systems.
5. Verify that an Environmental Contact is assigned by GRE who is responsible for: a. Correctly categorizing Category II Suppliers. b. Conducting Supplier Environmental Evaluations of Category II and Category III Suppliers (new and existing) consistent with requirements in the Guidance Document. c. Securing approvals, d. Entering required information into Supplier Evaluation EPD, e. Notifying CEA and Global Procurement when the result of a re-evaluation is unsatisfactory.

Guide note: Individuals performing Supplier Environmental Evaluations must possess the required skills and knowledge to evaluate the environmental attributes of the supplier and the services it provides, as well as the supplier facility when an on-site visit is required. Alternatively, the Business Organization can retain a qualified external consultant to perform the evaluation.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Chemical Management (EM008.P)

Objective:

To validate that chemical management programs and processes are in place and executed to comply with IBM and applicable legal requirements. Verify responsibilities have been identified, and procedures documented.

Approach:

1. Determine whether business organizations at the location purchase, develop, use, store, or distribute chemicals, including field use materials (FUMs) and chemical product supplies.
2. Has a chemical coordinator been identified who is responsible for providing coordinated chemical management to ensure programs are established and responsibilities assigned for identifying and complying with IBM and legal requirements? Verify that procedures and responsibilities have been documented to comply with IBM and applicable legal requirements for the storage, use, and distribution of chemicals.
3. Verify that resources and processes are in place to identify IBM and applicable legal requirements for chemicals used or distributed at the location or by the business organization.
4. Verify that procedures have been documented and responsibilities assigned to identify and comply with IBM and applicable legal requirements. Verify procedures address the following:
 - a. Identifying chemicals that will be managed as hazardous chemicals.
 - b. Obtaining or generating Safety Data Sheets (SDSs) and labels for hazardous chemicals.
 - c. Chemical hazard labeling.
 - d. Maintaining SDSs and chemical approvals in IBM's SDS Management System (Sphera).
 - e. Hazard Communication and employee training.
 - f. Review and approval of chemicals stored, used, and / or distributed.
 - g. Review and approval of chemical purchases including inter-plant transfer of chemicals, inter-plant transfer agreements, chemicals consigned to vendors or contractors, chemicals purchased for sale or use by customers or clients, and chemical samples.
 - h. Chemical receipt and chemical distribution.
 - i. Tracking chemicals that are unstable, highly toxic, or become hazardous with age.
 - j. Review and approval of contractor chemicals.
 - k. Management of chemical records according to requirements identified in the IBM Worldwide Records Management Plan (WWRMP).
5. For US locations, verify Tier II reporting under the US Superfund Amendments and Reauthorization Act (SARA) for chemical storage and location requirements are completed and filed within the regulatory deadline and that reports were reviewed with the appropriate levels of management. For non-US locations, verify that similar country-specific standards for reporting of chemical storage volumes and locations are met.

Guide Note: Review the annual SARA 313 Report (for US locations) and the SARA 311/312 report (for US locations). Check the incident log for any incidents with regulatory reporting notes and sample them to verify that notifications were addressed timely and that any follow-up action was completed on time or is being tracked.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Chemical Management (EM008.P)

(Continued)

Other Considerations:

- a. Report was filed late, but with permission from the regulatory authority.
 - b. Proper legal review was not completed.
 - c. Proper management review was not completed.
 - d. Required Follow-up Letters.
6. Select a sample of chemicals from the Tier II or similar country-specific report. During site tours test that the storage locations are accurate.
7. Identify any IBM core technology chemicals used on site and select a representative sample of those chemicals. Verify that all chemical in their formulations are disclosed by the chemical manufacturer / supplier. Verify that the recommended chemical information data set for the IBM upstream chemical review process has been obtained for these chemicals. Where exceptions occur, validate the exceptions are approved by the local chemical review team and CEA.
- Guide Note: CI ENV 203 contains a definition of IBM Core Technology: Methods, materials, chemicals or processes like fabrication, manufacturing, cleaning, or testing which are applied to the production/fulfillment of hardware products, the development of specifically engineered or synthesized chemicals or materials, or the design of proprietary products or processes. Examples include processes for layering, patterning, doping and heat-treating steps on a polished wafer; synthesis of multifunctional polymer micelles and/or dendrimers for co-delivery of therapeutic agents; and engineered nanomaterials.
8. Review the list of chemicals approved during the past six months. Select a representative sample and verify that the review included an assessment of potential health, safety, and environmental impacts where applicable as well as requirements for storage, use, and distribution. Verify that banned or restricted chemicals are not in use, or the use complies with IBM and legal requirements. Verify that SDSs and chemical approvals are being maintained in IBM's SDS Management System (Sphera) according to requirements identified in CI ENV203.
9. Obtain a listing of departments on the location that use chemicals, select a sample, including one contractor and one tenant if applicable, and perform a physical inspection of their work areas, e.g., chemical distribution center, chemical stockroom, chemical storage, etc., to assess compliance to the following requirements:
- a. 100% of the chemicals were reviewed and properly approved.
 - b. Approved chemicals in stock did not exceed maximum authorized quantities.
 - c. Chemicals were properly labeled, segregated and stored.
 - d. Employee hazard communication training is complete and current.

Guide Note: CI ENV 203

URL: <https://w3.ibm.com/w3publisher/corporate-environmental-affairs/about/corporate-policy-and-instructions>

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Chemical Management (EM008.P)

(Continued)

10. Verify that SDSs and chemical hazard labels are available in a language understandable by chemical-using employees at the location. Sphera updates the SDSs.

Guide Note: CI ENV 203

URL: <https://w3.ibm.com/w3publisher/corporate-environmental-affairs/about/corporate-policy-and-instructions>

11. Review a representative sample of chemical storage tanks, chemical piping, chemical containing process tools / equipment, and chemical waste containers to ensure they are properly labeled according to requirements identified in CI ENV 205.
12. Verify that the owner for the local Hazard Communication program and the organization(s) responsible for executing hazard communication program elements have been identified. Identify where the hazard communication program is documented.
13. Verify that chemical management records are being managed according to requirements listed in the WWRMP including:
- a. Hazard Communication training records (per WWRMP section MD008)
 - b. SDSs and chemical approvals not loaded in IBM's SDS Management System (Sphera)
 - c. Hazard determination for IBM-generated SDSs.
14. Select a representative sample of chemical-using departments and verify the department employees understand the location's chemical hazard labeling program, how to obtain an SDS and list of department chemicals, and how to obtain procedures and documents that comprise the hazard communication program. (Part of hazard communication questions.)
15. Verify that the location has a program to provide contractors with SDSs where required, and to explain the chemical hazard labeling program to contractors working on-site. (Part of hazard communication questions.)
16. Select a representative sample of chemicals that are unstable, highly toxic, or become hazardous with age, and verify the following:
- a. Unstable and highly toxic chemicals are secured to prevent unauthorized access.
 - b. Chemicals that become hazardous with age are tracked and disposed of prior to the expiration date (shelf-life date).
17. Conduct a review of tenant operations and verify the following:
- a. The tenant(s) has provided IBM with the list of hazardous chemicals used by tenant.
 - b. SDSs for tenant hazardous chemicals are available to IBM.
 - c. The tenant(s) is complying with the chemical management requirements identified in the lease agreement.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: TSCA (US Only) (EM009.P)

Note: Chemicals in Commerce and TSCA Program (US Only).

Objective:

To validate that programs and processes are in place and executed to comply with, the Toxic Substances Control Act (TSCA) in the US which regulates the introduction of new or already existing chemicals. Verify responsibilities have been identified, and procedures documented.

Approach:

1. Verify a location or business organization coordinator for the TSCA or country equivalent regulations has been identified and is clearly denoted in the local site chemical management processes.
2. Verify the location's written procedures for confirming the TSCA inventory status of all chemicals, mixtures, and chemical products that are manufactured, purchased, or imported by the site. Select a representative sample of new chemical approvals issued during the audit period and verify that the chemicals and/or ingredients of the chemical mixtures and products were reviewed for TSCA inventory status according to the site's written procedures.
3. Does the location manufacture or import chemicals? If yes, verify that the site has a documented procedure for assessing chemicals for TSCA or IBM import requirements or restrictions under Sections 4, 5, e.g., Pre-Manufacture Notice (PMN), Significant New Use Rules (SNUR), 6, and 7 of TSCA prior to commencing the manufacture or import of the chemical for any purpose. Importers of chemicals are required under TSCA Section 13 to certify that chemicals in shipments are either in compliance with TSCA or not subject to TSCA prior to entry. Ensure review of chemicals mixtures, and chemical products intended for import into the US by the site. Review records for the last 10 or 10% of the site's imported chemicals, mixtures, and chemical products (whichever is greater) over the past 6 months. Verify that there is a copy of the import certificate on file for those chemicals, mixtures, and chemical products subject to TSCA regulation.
4. Obtain a list of chemicals imported to the site during the audit period. Select a sample and review the import records for the site's imported chemicals, mixtures, and chemical products. Verify that there is a copy of the import certificate on file for those chemicals, mixtures, and chemical products subject to TSCA regulation. Verify also that the following information was confirmed either by the site TSCA Coordinator, or by a supplier warranty statement obtained by the site TSCA Coordinator, for every ingredient in the chemicals, mixtures, and/or chemical products prior to the issuance of the import certification:
 - a. The chemical is on the TSCA Inventory or has an IBM-held exemption from PMN.
 - b. The chemical is not subject to a TSCA Section 4 test rule.
 - c. The chemical is not subject to import prohibitions under TSCA Sections 5(a)(2), 6, or 7 rules or actions and / or that the product was imported under a TSCA Research and Development (R&D) Exemption.
5. Does the site manufacture or import new chemical substances, i.e., substances that are not listed on the TSCA inventory, exclusively for R&D applications under the TSCA R&D Exemption from PMN? If so, verify the site has a documented and implemented TSCA R&D Exemption process and verify that the proper records substantiating the R&D exemptions from PMNs for non-IBM and IBM chemicals are being retained.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: TSCA (US Only) (EM009.P)

(Continued)

6. Does the location support R&D and/or commercial manufacturing activities at the facilities? If so, verify that the site has a documented process for tracking, handling, and disposing of R&D Use-Only chemicals. Request the inventory of R&D Use-Only chemicals currently on-site. Review a representative sample of the record files of these chemicals and track their use and movement at the site. Verify that the R&D Use-Only chemicals are not being used for IBM commercial applications or other applications besides R&D. Verify that the containers clearly state that the substance is for R&D Use-Only.
7. Verify the location of the site's documented procedure for Section 8(c) on the environmental allegations recording rule. Verify that the site's documented procedures include mechanisms to ensure that the site TSCA Coordinator is notified of all environmental allegations that are made to the site within 24 hours of the allegation being received at the site. Verify that the site has a TSCA Section 8(c) environmental allegations notebook that contains records of all Section 8(c) recordable environmental allegations made at that site.
8. Verify the location of the site's documented procedures for reporting substantial new risk information under Section 8(e). Verify that the documented procedures include a mechanism that ensures that the site TSCA Coordinator is informed immediately of any new information on environmental effects or environmental contamination associated with the site. Review records for all environmental incidents that occurred at the site and findings from all environmental and / or toxicology studies initiated during the audit period. Verify that the site procedure was implemented and that all environmental incidents and studies were evaluated by the site TSCA Coordinator for possible reporting under Section 8(e).
9. Does the location export chemicals? Verify the location of the site's documented procedures for Section 12(b) export notification requirements. Review the records for all chemical exports initiated by the site within the last 12 months. Verify that all chemical products exported from the site were reviewed for export notification requirements prior to being exported.
10. Verify that the location has retained documentation of the review conducted of the site's chemical management records to determine if the site manufactured (including by-products) and/or imported any chemicals in volumes sufficient to trigger reporting for TSCA Chemical Data Reporting requirements.
11. TSCA Section 13 Import Certification. Identify the location of the site's documented procedures for reviewing chemicals mixtures, and chemical products intended for import into the US by the site. Review records for the last 10 or 10% of the site's imported chemicals, mixtures, and chemical products (whichever is greater) over the past 6 months. Verify that there is a copy of the import certificate on file for those chemicals, mixtures, and chemical products subject to TSCA regulation.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: TSCA (US Only) (EM009.P)

(Continued)

12. For the chemicals, mixtures, and/or chemical products identified in question 11, verify that the following information was confirmed (either by the site TSCA Coordinator or by a supplier warranty statement obtained by the site TSCA Coordinator) for every ingredient in the chemicals, mixtures, and/or chemical products prior to the issuance of the import certification: -the chemical is on the TSCA Inventory or has an IBM-held exemption from PMN; and -the chemical is not subject to a TSCA Section 4 test rule; and -the chemical is not subject to import prohibitions under TSCA Sections 5(a)(2), 6, or 7 rules or actions and / or that the product was imported under a TSCA R&D Exemption.
13. Importers of chemicals are required under TSCA Section 13 to certify that chemicals in shipments are either in compliance with TSCA or not subject to TSCA prior to entry.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Air Emissions (EM010.P)

Objective:

To verify that the location process will result in compliance with IBM and legal (including permit) requirements for Air Emissions. To verify that all air emissions units have been identified and that potential off-site consequences associated with routine discharges of hazardous air pollutants, volatile organic compounds, global warming gases, and boiler emissions have been characterized.

Approach:

1. Verify the location has a process for air emissions management consistent with legal and permit requirements and CI ENV 204 - Resource Conservation and Pollution Prevention.
2. Review applicable legal, including permit, requirements for the location. Review the site air emissions inventory and validate that it is complete and up to date. Confirm that there is a process for updating the air emissions inventory as a result of significant changes at the location. At a minimum, this should occur annually.
3. Validate that emission characterization methods are appropriate.
4. Review the location process to verify that samples are collected by qualified personnel and competent and / or accredited labs who performed any required environmental analyses. Verify that the process addresses methods for communicating requirements to IBM suppliers and validate that this communication was performed.
5. Validate that when instances of ambient air quality exceed benchmark concentrations in CI ENV 204 and risk assessments indicate a concern, appropriate steps are taken to decrease emissions.
6. Validate that any required emissions reduction plans and risk assessment documents are in place and submitted to CEA.
7. Verify that non-compliance issues and permit exceedances are reported to the appropriate government agencies when applicable, site management and CEA, per CI ENV 204, in a timely manner.
8. Verify that, when applicable, a root cause analysis was performed for noncompliance, and that corrective action plans were executed. Verify action plans included target completion dates and were monitored until compliance was achieved.
9. Select a representative sample of air emission units and verify compliance with IBM and legal, including permit, requirements.
10. Verify that where there is an exceedance and further evaluation of potential off-site impact indicates a concern, either an emission reduction plan has been established and implemented, or a risk assessment documented and approved by the location senior executive. Verify that the reduction plan or approved risk assessment was submitted to CEA.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Air Emissions (EM010.P)

(Continued)

11. As applicable, confirm that a documented schedule exists for required training for technicians handling air emissions monitoring, sampling, or testing. Validate that training requirements have been met in a timely manner.
12. Verify that the location has identified environmental monitoring equipment that requires periodic calibration for the purpose of validating and/or demonstrating compliance with IBM or legal, including permit, requirements. Validate that the list of equipment is maintained and current and that the periodic calibration was performed in a timely manner.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Water Discharges (EM011.P)

Objective:

To verify that the water discharge process is in compliance with IBM and legal, including permit, requirements. Confirm that the location has identified all water discharges. In addition, determine if appropriate controls are in place to identify chemicals that will be present in water discharges.

Approach:

1. Review applicable IBM and legal, including permit, requirements for the location.
2. Confirm that the inventory of water discharges was complete. Validate that there was a process in place to determine when updates to the inventory are required due to major process modifications at the location.
3. Verify that wastewater streams are properly characterized with regards to quantity, chemical composition, and physical properties. Select a sample of monitoring data reports from the audit period and verify compliance with the requirement that wastewater characterization properly represents the measured wastewater composition.
4. Test that appropriate monitoring has been performed to identify discharges and changes to discharges.
5. Review the location process to verify that samples are collected by qualified personnel and competent and/or accredited labs who performed any required environmental analyses. Verify that the process addresses methods for communicating requirements to IBM suppliers and validate that this communication was performed.
6. Select a representative sample of discharge reports for the audit period and verify compliance with legal requirements, including permit limits, and CI ENV 204 discharge limits.
7. When applicable, verify a root cause analysis was performed for non-compliance situations and that corrective actions plans were executed. Verify action plans included target completion dates and were monitored until compliance was achieved.
8. Verify that non-compliance issues and permit exceedances were reported to the appropriate government agencies when applicable, site management and CEA, per CI ENV 204, in a timely manner.
9. Select a sample of discharges and verify compliance with IBM and legal, including permit, requirements.
10. Verify that there is a documented schedule for any required training for technicians handling water discharge monitoring, sampling, or testing. Validate that training requirements have been met and in a timely manner.
11. Verify that the location has identified environmental monitoring equipment that requires periodic calibration for the purpose of validating and/or demonstrating compliance with IBM or legal, including permit, requirements. Validate that the list of equipment is maintained and current and that the periodic calibration was performed in a timely manner.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Hazardous and Nonhazardous Special Waste (EM012.P)

Objective:

Verify that the location has established a process for compliance with applicable IBM, legal, and permit requirements for management of hazardous and nonhazardous special waste. Verify all hazardous and nonhazardous special waste streams have been identified and adequately characterized to determine correct methods of handling, storage, and disposal.

Approach:

1. Verify that an inventory of hazardous and nonhazardous special waste streams has been generated at the location and a management plan has been developed.
2. Review applicable legal, including permit, requirements for the location. Review local procedures and visit a sample of areas involved in the generation, handling, and storage of hazardous and chemical wastes. Test the procedures to verify that the following attributes are functioning as intended:
 - a. All hazardous waste streams are identified and properly characterized based on analytical work or knowledge. Where the environmental group uses, documentation provided by manufacturing, service, or other organizations to classify wastes, check a representative sample of the documents generated during the audit period to verify compliance with requirements that:
 - i. The waste descriptions generated by the department provide information about the waste.
 - ii. That they correlate to the correct location of the waste stream.
 - iii. That all documentation was up to date for the audit period.
 - b. Collection and handling methods are appropriate, and storage areas have appropriate secondary containment.
 - c. Waste containers are appropriately marked, stored, packaged, and segregated.
 - d. Waste is collected only in designated areas which were clearly labeled.
 - e. Treatment and disposal methods were appropriate.
 - f. Access to waste disposal areas was appropriately restricted.
 - g. Hazardous waste was disposed-off in a timely manner, rather than allowing excess waste to accumulate at the site.
 - h. Review the location process to verify that samples are collected by qualified personnel and competent and/or accredited labs who performed any required environmental analyses. Verify that the process addresses methods for communicating requirement to IBM supplier and validate that this communication was performed.
 - i. As applicable, root cause analysis is performed for non-compliance situations, and corrective action plans executed. Action plans included target completion dates and are monitored until compliance is achieved.
 - j. Non-compliance issues and permit exceedances are reported to both the appropriate government agencies and CEA in a timely manner.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Hazardous and Nonhazardous Special Waste (EM012.P)

(Continued)

3. Select a sample of hazardous waste streams and verify compliance with applicable legal, including permit, requirements.
4. Where hazardous waste inspections are required by IBM location or legal requirements, check inspection documentation for the audit period. Verify that documentation meets the inspection criteria regarding scope, schedule, frequency, and other inspection requirements mandated by permit or local procedures. Verify that all identified problems are corrected, and corrections are noted in the inspection log.
5. Select a sample of hazardous waste shipments and test the following attributes:
 - a. Required shipping documents, e.g., US waste manifest or country equivalent document, were completed and accurate.
 - b. Hazardous and Nonhazardous Special waste disposal vendors were approved by IBM CEA per CI ENV 205 for the specific waste sent to them.
6. Verify that there is a documented schedule for required training for personnel handling hazardous waste. Validate that training requirements have been met on a timely basis.
7. Verify that the location has identified environmental monitoring equipment that requires periodic calibration for the purpose of validating and / or demonstrating compliance with IBM or legal, including permit, requirements. Confirm that the list of equipment, if applicable, is maintained and current and that the periodic calibration was performed in a timely manner.
8. Review the location's waste management program and validate that it is current and being followed, and that it includes the following with respect to achieving objectives and targets:
 - a. The time frame for achieving objectives and targets.
 - b. The person responsible.
 - c. The means for achieving objectives and targets.
9. Verify that, if the targets and objectives of the hazardous waste management program are not met, that there is a plan in place to improve performance.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Nonhazardous Waste (EM013.P)

Objective:

To verify that the process for the management of nonhazardous wastes at the site complies with IBM and legal requirements. To determine if the location has established nonhazardous waste recycling objectives and targets to support the objectives and targets established by corporate.

Approach:

1. Review legal requirements and the site non-hazardous waste management plan, including provisions relative to IBM's nonhazardous recycling goals. Verify that there is a plan to meet or ensure best possible performance against corporate recycling goals for nonhazardous waste.

Guide Note: CI ENV 204: Each site must understand its legal requirements pertaining to proper waste management. IBM's Nonhazardous Waste goal applies to (a) IBM-managed locations and (b) leased locations greater than 100,000 square feet, where practical, to be recycled.
2. Verify that an inventory of nonhazardous waste streams generated by IBM business operations at the location have been completed and a nonhazardous waste management plan developed. Verify action plans are in place to complete this inventory and plan if not currently available.
3. Verify that a documented schedule exists for any required training for technicians handling nonhazardous waste. Validate that training requirements have been met and in a timely manner.
4. Check the Location's Central Storage Area for Wastes, the main area for collecting recyclables, and sample areas used for the accumulation of waste and verify compliance with following conditions:
 - a. The number and types of containers in the area match the inventory or accumulation logbook where such document is required.
 - b. All containers are labeled per IBM and / or legal requirements.
 - c. All containers are segregated and closed as required by IBM and / or legal requirements.
 - d. Where required, the containers are associated with an internal waste document / description.
 - e. Containers moved from on-site areas to inventory or inventory to treatment and disposal areas, or facilities were moved in accordance with location and / or legal requirements.
 - f. Where required, all accumulation and storage areas are clearly marked.
5. Verify that the location has identified environmental monitoring equipment that is to be calibrated periodically for the purpose of validating and demonstrating compliance with IBM or legal, including permit, requirements. Validate that the equipment was calibrated in a timely manner and the list of equipment is current.
6. Review the location's waste management program and validate that it is current and being followed, and that it includes the following with respect to achieving objectives and targets:
 - a. The time frame for achieving objectives and targets.
 - b. The person responsible.
 - c. The means for achieving objectives and targets.
7. Verify that, if targets and objectives of nonhazardous waste management program are not met, there is a plan in place to improve performance.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Containment Systems, Groundwater Programs, and Environmental Remediation Activities (EM014.P)

Objective:

To verify that the location has the necessary controls in place for secondary containment systems and is accounting for the costs associated with environmental remediation activities in accordance with IBM accounting practices.

Approach:

Review the location's secondary containment systems and the accounting process for the identification and recording of costs associated with environmental remediation activities and corrective actions. Verify that the following attributes are in place:

1. For IBM-owned or managed location(s), where IBM has responsibility to manage facility operations including emergency backup systems (generators, battery banks, etc.) or other liquid and Liquid Storage and Handling Systems (as defined in CI ENV 204), verify a process is in place to identify and ensure secondary containment is in place per legal and IBM CI ENV 204 requirements. <https://w3.ibm.com/w3publisher/corporate-environmental-affairs/about/corporate-policy-and-instructions>
2. Verify that the location maintains a current inventory of liquid storage and handling systems non-compliant with secondary containment requirements of CI ENV 204.
3. Verify a procedure is in place for tank inspection and testing for verification of continued integrity. Verify that appropriate means are specified in the procedure.
4. Request an inventory of all containment areas, leak detection alarm and pump out systems, and containment valves, and verify there is a preventive maintenance (PM) and inspection schedule for these areas. Verify that all areas, detection systems, and control valves are on the PM system and scheduled PM activities were completed per requirements. Inspect a representative sample of the installations to ensure they are in an acceptable condition.
5. Verify that elevator hydraulic oil reservoirs and pit floors are inspected and maintained in accordance with maintenance schedules to prevent accidental releases. Verify that the required PM is on the PM system. Verify that accurate and repeatable accounting of hydraulic fluid is being performed on regular basis.
6. Verify that systems that are not in compliance with the secondary containment requirements of CI ENV 204 have been reported to CEA and entered in the environmental performance database per requirements of CI ENV 204.
7. Verify that there is a process in place for locations to report, annually, a summary of chemical releases to "Engineered Secondary Containment Systems" via the Environmental Performance Database (EPD).

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Containment Systems, Groundwater Programs, and Environmental Remediation Activities (EM014.P)

(Continued)

8. Verify the location has assessed requirements of CI ENV 204 Groundwater and Soil Protection to determine if they are applicable to the location. Where applicable, a potential release evaluation must have been completed and be available for review. Verify that locations have complied with the following requirements of CI ENV 204:
 - a. Verify that a potential release evaluation has been completed, if applicable.
 - b. If the site assessment suggested the possible presence of contamination, verify that an investigation been initiated. Verify that any intrusive investigations had CEA involvement and used a CEA -approved environmental consultant.
 - c. Verify that, if necessary, actions have been initiated or a plan is in place to protect human health, or if the source is on IBM property, to prevent off-site migration. Verify that the requirement of CEA involvement and the use of a CEA-approved consultant was met.
 - d. Verify that a site required to perform groundwater monitoring has a Groundwater Monitoring Plan and that it contains the required elements.
 - e. Verify that sites with groundwater monitoring wells have a process in place to implement the required Groundwater Monitoring Plan elements.
 - f. If existing wells are no longer utilized as part of the Groundwater Monitoring Plan, verify a plan is in place to properly abandon the wells in accordance with local environmental requirements.
 - g. Review the previous year's monitoring results and verify that the required elements are being implemented and that a process is in place for the review of data by a qualified person.
9. For locations performing environmental remediation, verify:
 - a. Annual costs associated with environmental remediation activities and corrective actions are appropriately recorded in the accounting ledger.
 - b. Cost projections for the duration of the environmental remediation activities and corrective action plans have been properly estimated and entered into the CEA's Environmental Performance Database and corporate balance sheet accounting (at least annually but not later than February 15 of each year for the prior year).
 - c. Costs of environmental remediation activities and corrective actions are periodically (at least annually) monitored and reviewed with appropriate levels of location management.
 - d. CEA had been advised of these environmental remediation activities at least annually.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Laboratory QA/QC (EM015.P)

Objective:

Where IBM uses results of emissions, e.g., air, wastewater, waste, etc., sampling and laboratory analyses to demonstrate environmental performance, and compliance with legal or IBM requirements, the labs performing sampling and analysis must have Quality Assurance (QA) plans and programs to ensure accurate laboratory results are used for this purpose. This section will determine if effective processes exist to ensure laboratory QA and Quality Control (QC) manuals are current for all analytical laboratories (IBM and vendor laboratories) and to validate completeness of QA / QC controls and trend monitoring as required in the QA / QC manual.

Approach:

Verify that laboratory QA and QC manuals are kept current for all on-site, i.e., IBM-owned, analytical laboratories. Test to validate completeness of QA / QC controls and trend monitoring as required in the QA / QC manual.

1. If the location is using only certified or accredited vendor laboratories, verify that vendor laboratories provide, on an annual basis, a certification assuring IBM that they completed proper and documented QA / QC controls and trend monitoring as required by the vendor's own QA / QC procedures.
2. Verify periodic audits of vendor laboratories were conducted to verify that the required instrument calibrations are taking place, that problems and solutions are recorded in an equipment logbook, that the control charts are being built, and that deviations from the control plan are addressed and closed.
3. Verify the equipment logbook documents the completion of the maintenance activity.

Guide Note: Where IBM uses results of sampling and laboratory analyses to demonstrate environmental performance, legal compliance, and IBM requirements, it is critical the laboratories performing the sampling and analysis services have QA plans and programs. IBM requires annual certification from these service providers regarding the following aspects of their programs which should be verified during vendor audits:

1. Periodic calibration requirement for specific instruments: identify which equipment requires periodic calibration (pH probes, ammonia probes, Atomic Absorption Spectrophotometers, Gas Chromatography, etc.) and that calibrations were performed.
2. Equipment should have control charts utilizing the analysis of known standards.
3. Specified maintenance requirements for equipment specified in IBM WW EMS Manual 9.1.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Monitoring Equipment (EM016.P)

Objective:

To determine if effective processes exist to ensure that monitoring equipment used for the purpose of validating and/or demonstrating compliance with IBM and/or applicable legal requirements are properly calibrated.

Approach:

Verify monitoring equipment, e.g., flow meters, pressure cells, thermometers, scales, on-line pH meters, flow proportioned composite samples, on-line air emission or wastewater monitors, and other measuring equipment, etc., used for validating and/or demonstrating compliance with IBM or applicable legal requirements are calibrated.

Guide Note: Equipment whose purpose does not consist of environmental compliance monitoring, e.g., thermometers for temperature sensing in a manufacturing area, flow meters for measuring air flow through a clean room), etc., are not covered by this question. A documented program requiring equipment used for gathering samples or data for legal reporting and / or compliance demonstration undergo scheduled calibration and validation of the readings must be in place. At a minimum, this program shall comply with applicable legal, including permit, requirements per IBM WW EMS Manual 9.1.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Training and Communications (EM017.P)

Objective:

To determine if effective processes exist to train and communicate with IBM employees and contractors. To verify that procedures for communicating to internal and external parties about the EMS are established and implemented at the location.

Approach:

1. Verify that training needs have been identified for all environmental programs and that an effective training program has been documented and implemented to meet these needs.
2. Verify that the location has a documented schedule for required environmental, energy, and chemical management training of IBM employees.
3. Verify that on-site contractors and subcontractors providing environmental and chemical management services have a documented schedule for required training. Verify that the location and/or the contractors and subcontractors are in compliance with the established requirement that their respective employees for whom training is required have the current training and that training was completed prior to initiation of job responsibilities or within specified time frame.
4. Review the location's training program and verify that it meets the following attributes:
 - a. Training procedures and records are documented.
 - b. All employees and on-site contractors performing tasks which can cause a breach of legal compliance or significant environmental impacts received appropriate training and are competent. In countries where specific training requirements are prescribed by legal requirements, ensure such requirements are met, e.g., include hazardous waste technician training specified by RCRA and the OSHA HAZWOPER training in the US, etc.
 - c. Appropriate training should include:
 - i. The importance of conformance to environmental procedures.
 - ii. Significant environmental impacts related to their work activities.
 - iii. Environmental benefits of improved personal performance.
 - iv. Roles and responsibilities regarding environmental practices and emergency preparedness and response.
 - v. Potential consequences of departures from procedures.
 - vi. Consideration of individual's competencies, based on education, training, and experience.
5. Select a representative sample of employees and on-site contractors and test to confirm that training has been completed. If there are government or location training requirements for nonhazardous waste collection workers, include in the above sample a selection of the identified workers and verify that their training is current.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Training and Communications (EM017.P)

(Continued)

6. Verify the location has a documented process to communicate environmental information to management and the general location population, including on-site contractors and subcontractors as appropriate. Request to see specific communications completed for the past six months and verify the following had been communicated to IBM employees and on-site contractors and subcontractors providing environmental and chemical management services:
 - a. Corporate Policy # 139.
 - b. The importance of conformance to environmental procedures.
 - c. Significant environmental impacts related to their work activities.
 - d. Environmental benefits of improved personal performance.
 - e. Their roles and responsibilities regarding environmental practices and emergency preparedness and response.
 - f. Potential consequences of departures from procedures.
7. Select a sample of communications and validate their accuracy, timeliness, completeness, and accessibility to employees and on-site contractors.
8. Verify that procedures address methods for communicating applicable requirements to IBM suppliers and validate that this communication was performed.
9. Validate that there is a procedure for:
 - a. Internal communications, at all relevant levels, within the organization and between the location and CEA.
 - b. Receiving and responding to relevant communication from external interested parties.
Test that the procedure was implemented.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Management of On-Site Environmental Contractors (EM018.P)

Objective:

To verify that contractors providing IBM with on-site services related to the environmental management are performing in accordance IBM requirements and the stated scope of work.

Approach:

1. Review appropriate programs including, but not limited to, management of IBM's environmental affairs goals, chemical management, legal and permit compliance, water treatment, etc.:
 - a. Validate there is a documented process to communicate IBM requirements, and changes to the requirements, to contractors and subcontractors providing environmental, energy and chemical management services.
 - b. Confirm that IBM requirements have been communicated to contractors and subcontractors.
 - c. Validate contractors and subcontractors have implemented the IBM requirements.
2. Verify that on-site contractors and subcontractors have documented processes requiring the strict adherence to IBM requirements.
3. Verify procedures have been established for the on-going monitoring of contractor and subcontractor performance to confirm adherence to IBM requirements.
4. Verify a formal process has been established to document vendor deficiencies in the performance of the scope of work to IBM requirements, and corrective action plans have been implemented as appropriate.

Guide Note: CI ENV 201: Contract and Statement of Work for the contracted services. A contractor's failure to perform in accordance with IBM's requirements could result in unwanted environmental incidents, non-compliance with legal requirements, and/or damage to IBM's image. When an environmental incident occurred and even if the on-site contractor providing the service took full responsibility, the public and the government would likely still consider IBM ultimately responsible, e.g., any fines or legal proceedings would likely be levied against IBM, not the contractor. Examples of services on-site contractors provide to IBM include operation of waste-water treatment plant, those related to significant energy uses, chemical operations, preparation of legal filings, preparation of waste manifests, etc.

URL Name: CI ENV 201

URL: <https://w3.ibm.com/w3publisher/corporate-environmental-affairs/about/corporate-policy-and-instructions>

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Evaluation of Tenants (EM019.P)

Objective:

To determine if a process exists to validate those tenants in IBM-owned or leased facilities have implemented environmental and chemical management processes which comply with IBM requirements, as defined in the lease and legal requirements, and which minimizes potential financial and legal liabilities to IBM.

Approach:

1. Review the lease between IBM and any tenants occupying IBM owned or leased facilities. Identify the environmental and chemical management requirements specified in the lease agreement. Validate that the following were incorporated in the terms and conditions of the lease:
 - a. Requirements for tenant compliance with all applicable IBM and legal requirements.
 - b. Requirements that the tenant provide IBM with appropriate measurements and information to demonstrate compliance with IBM and legal requirements.
 - c. IBM's rights to perform periodic audits of the tenants' operations to verify compliance with IBM and legal requirements.
 - d. A process for the implementation of corrective action plans to address any identified tenant environmental exposures. The process must include formal communication of the completion of the corrective actions.
 - e. If the lease does not address an item identified above, a documented recommendation must be made to the appropriate IBM management concerning the incorporation of the specific item.
2. Review the tenant's environmental and chemical management processes. Validate that processes have been implemented per the terms and conditions of the lease for the following:
 - a. Environmental permit compliance.
 - b. Maintenance of a list of chemicals used in tenant operations (including fuel storage and emergency generators).
 - c. SDS availability.
 - d. Environmental incident prevention and reporting.
 - e. Hazardous waste management.
 - f. Non-hazardous waste management.
3. Select a sample of the tenant occupied areas and operations for review. Verify that the following process elements are effectively implemented per the terms and conditions of the lease:
 - a. Environmental permit requirements are identified and complied with.
 - b. Chemical list has been provided to IBM and reviewed against IBM's banned and restricted chemical listing, as well as, national and local regulatory requirements.
 - c. Secondary containment requirements are identified and complied with.
 - d. Safety Data Sheets (SDSs) are available for tenant chemicals.
 - e. Hazardous waste is appropriately accumulated, stored and disposed of.
 - f. Non-hazardous waste management and recycling programs have been implemented.
 - g. Environmental incidents are identified, documented, and communicated to IBM and regulatory agencies, as applicable.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Environmental Due Diligence for Business Transactions (EM020.P)

Objective:

Verify effective processes exist for completing the appropriate level of environmental due diligence for business transactions per CI ENV 207. Business transactions (with or without involving real estate) include acquisitions and divestitures, and leases of real estate.

Approach:

1. Verify there is a process in place to ensure appropriate level of environmental due diligence in conformance with CI ENV 207 for transactions involving real estate (acquisitions, divestitures, lease or sub-lease engagement or termination, etc.). Verify the process includes the requirement to engage CEA prior to initiating any environmental site assessment. Verify this process has been communicated to affected personnel within the organization.
2. Select a representative sample of transactions completed during the audit period and verify that CEA has been engaged on a timely basis to initiate environmental site assessments. Ensure that environmental matters in the terms and conditions have been reviewed by CEA's legal counsel.
3. Verify that the environmental site assessment final report for transactions by which IBM acquired or terminated an interest in real estate was sent to CEA.
4. Verify there is a process in place to ensure CEA reviews the environmental terms and conditions between IBM and non-IBM tenants. Verify there is a process in place to ensure non-IBM tenants are aware of and follow applicable environmental requirements.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Reporting into CEA Environmental Performance Database (EM021.P)

Objective:

To verify that the information provided to CEA in the Environmental Performance Database (EPD) and Cority was accurate and has been prepared and submitted in accordance with the criteria prescribed by CEA.

Approach:

Review a sampling of the EPD reporting for compliance with corporate requirements regarding the following attributes:

1. Based on the primary tests in the audit program, verify that the EPD accurately reflected the location's environmental programs.
2. Verify that the Business Organization or IBM location submitted required forms to CEA in the Environmental Performance Database (EPD) on time (on or before the last business day in February of the current year for the previous reporting year for annual reporting), including quarterly submission requirements for certain data.

Guide Note: Required quarterly submissions include hazardous waste reduction, nonhazardous waste recycling, and product end-of-life management. Verify a sample of the hazardous waste, wastewater, and air emissions EPD data against regulatory reports to ensure accuracy. Review other data, e.g., water consumption and conservation, and verify they are accurate.

URL Name: CI ENV 201.

URL: <https://w3.ibm.com/w3publisher/corporate-environmental-affairs/about/corporate-policy-and-instructions>

3. Verify the EPD was reviewed by appropriate levels of management. Verify that a note confirming the responsible executive review and approval of annual reporting was sent to the director of CEA prior to form submittal.

Guide Note: CEA requires an email from the responsible executive confirming his/her review and approval of the EPD.

4. Verify all required forms of the EPD or annual report were completed.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Documentation and Records Control (EM022.P)

Objective:

To determine if the process for documentation and records control is effectively managed.

Approach:

Review procedures for documentation management. Examine a sample of documentation and records.

Validate that the following attributes have been executed:

1. Procedures have been documented for all activities that could cause a breach in legal requirements or significant environmental impact. Verify that the location had a procedure to identify and control relevant documentation. Test the documentation to confirm the following:
 - a. It was properly approved.
 - b. It was current, dated (with revision dates available), and obsolete documents were removed.
 - c. It was legible, and identifiable.
 - d. It was maintained in an orderly manner.
2. Records needed to manage environmental programs are retained consistent with the Worldwide Records Management Plan (WWRMP) document or site-specific requirements for records not mentioned in WWRMP. The minimum required records include training records, the results of environmental management system audits, and management reviews. Determine whether the location has any special records management requirements detailed in appendices to the WWRMP and verify that the location has determined what data needs to be controlled to satisfy the additional requirements, that the necessary data collection process is in place to collect the required data, and that periodic reviews are completed to verify that the data is being collected as required.

Guide Note: IBM Worldwide Records Management Plan (<http://w3-03.ibm.com/ibm/documents/corpdocweb.nsf/ContentDocsByTitle/Worldwide+Records+Management>) and the Environmental Management System manual. In addition to corporate records retention requirements, there may be legal, or location established requirements. Where conflicts exist, the more stringent of the requirements shall be followed. The location should have a process to collect and maintain all critical documentation pertinent to an incident or inspection in a central, easily accessible location.

URL Name: IBM Worldwide Records Management Plan

URL: ([http://w3-](http://w3-03.ibm.com/ibm/documents/corpdocweb.nsf/ContentDocsByTitle/Worldwide+Records+Management)

[03.ibm.com/ibm/documents/corpdocweb.nsf/ContentDocsByTitle/Worldwide+Records+Management](http://w3-03.ibm.com/ibm/documents/corpdocweb.nsf/ContentDocsByTitle/Worldwide+Records+Management))

3. Records were protected against damage, deterioration, and loss.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Monitoring, Measurement, Assessment, and Management Review (EM023.P)

Objective:

To verify that an effective process exists to monitor the performance of environmental programs, and that the required self-assessments and reviews are accurately performed.

Approach:

1. Review the site's environmental measurements process. At a minimum, measurements should include non-hazardous waste recycling and energy conservation. The site may designate additional measurements and objectives. Verify the following attributes:
 - a. Test a sample of source documentation to verify environmental measurement data is accurate.
 - b. Performance is monitored against established objectives.
 - c. Responsibility for meeting each objective has been identified.
 - d. Objectives not met have resulted in root-cause analyses and corrective action plans, and target dates were assigned for achieving each objective.
 - e. Measurements are reported periodically to the appropriate levels of management.
2. Review the location procedures for conducting self-assessments of the environmental management system and validate the following attributes:
 - a. At a minimum, an annual self-assessment has been completed using the IBM developed self-assessment protocol.
 - b. Self-assessment against applicable legal requirements, i.e., assessment of legal compliance.
 - c. The self-assessment involves the actual testing of relevant field operations. Request test documentation and verify that sample documentation of the tests is available and that the reported test results are supported by the documentation.
 - d. The self-assessment scope has been documented and included all requirements of the WW EMS Manual and the location's documented environmental management system.
 - e. Results are reported to appropriate levels of management and used for Business Controls Quarterly Certification, as applicable.
 - f. Root-cause analyses and corrective action plans are executed in a timely manner for deficiencies found during self-assessments. Action plans include target completion dates and are tracked to closure.
3. Take a sample of audit and self-assessment findings (including Corporate Audit, EMS internal audit, ISO 14001 audit, ISO 50001 audits and professional self-assessments, etc.) generated during the audit period. Verify that:
 - a. All identified deficiencies were communicated to the area owner.
 - b. A response was received for each identified deficiency.
 - c. All action items were tracked to closure.

ENVIRONMENTAL MANAGEMENT – CHEMICAL USING MDR LOCATIONS AUDIT PROGRAM #47-10-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Environmental Management Programs

Primary Test: Monitoring, Measurement, Assessment, and Management Review (EM023.P)

(Continued)

4. Verify that results of self-assessment (self-assessment rating) have been reported to appropriate levels of management and the organization's Business Controls.
5. Verify that the Professional Self-Assessment (PSA) was completed in a timely manner and entered in the CEA-designated database as required.
6. Review the location procedures for conducting EMS audits and validate the following attributes:
 - a. EMS audits are completed as per the requirement in the WW EMS Manual - Section 9.2 at least annually.
 - b. EMS audit results are reported to appropriate levels of management.
 - c. EMS audits are conducted by auditors who can act impartially and objectively when reviewing the EMS programs.
 - d. EMS audits are performed by personnel who have the proper experience and training.
 - e. Root-cause analyses and corrective action plans are executed for deficiencies found during EMS audits. Action plans include target completion dates and are tracked to closure.
7. Verify that the location reassessed the applicability of the Professional Self-Assessment for items previously deemed non-applicable at least once every year.

Guide Note: Changes to a location's mission, legal and / or IBM requirements could result in changes in applicability of certain requirements. It is critical to the integrity of the Environmental Affairs self-assessment program that locations work from a current and accurate set of questionnaires.

8. Validate that management reviews are performed, and that adequate data is presented for management to determine the suitability, adequacy, and effectiveness of the EMS. Confirm that reviews cover the following:
 - a. Status of actions from previous management reviews.
 - b. The extent to which relevant objectives and targets have been met.
 - c. Changing circumstances that materially impact the implementation of the EMS, including changes to activities, products, and services, needs and expectations of interested parties - including compliance obligations, significant environmental aspects, and risks and opportunities.
 - d. Environmental performance, including results of the internal EMS audit(s), status of nonconformities and corrective actions, and fulfilment of compliance obligations.
 - e. Adequacy of resources for maintaining the effectiveness of the relevant EMS.
 - f. Relevant communication(s) from external interested parties, including complaints.
 - g. Opportunities for continual improvement.
9. Verify that the senior location executive or other top management representative participates in these reviews.